

UNIT 4: Polymers, Organometallics & Composites

1. Classification of Polymers

Polymers are giant macromolecules constructed by linking small repeating chemical units (monomers).

Thermoplastics vs Thermosetting Polymers:

- Thermoplastics: Linear or branched polymers with weak van der Waals forces. Soften on heating, easily reshaped & recycled (e.g., Polyethylene, PVC, Polystyrene).
- Thermosets: Highly cross-linked 3D networks. Do not soften on heating once molded; decompose under high heat (e.g., Bakelite, Vulcanized rubber, Epoxy resin).

Conducting Polymers:

Polymers that conduct electricity due to continuous conjugated π -bonds (alternating single and double bonds).

Examples: Polyacetylene $(-\text{CH}=\text{CH}-)_n$, Polyaniline.

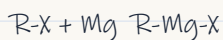
Doping Concept: Conductivity increases by million times upon oxidation (p-doping with I_2) or reduction (n-doping with Na). Applications: OLED screens, solar cells, biosensors.

2. Organometallic Compounds

Organometallics contain at least one direct Carbon-Metal (C-M) bond.

Grignard Reagents (R-Mg-X):

Prepared by reacting alkyl halide with Mg ribbon in dry ether:



Reactivity: C-Mg bond is strongly polar ($\text{C}^{\delta-} - \text{Mg}^{\delta+}\text{X}$). Acts as a strong nucleophile.

- Reaction with HCHO Primary Alcohol.
- Reaction with R-CHO Secondary Alcohol.
- Reaction with Ketones Tertiary Alcohol.
- Reaction with CO_2 Carboxylic Acid.

3. Industrial Organometallic Catalysts

Ziegler-Natta Catalyst:

Combination of Triethylaluminium and Titanium tetrachloride $[Al(C_2H_5)_3 + TiCl_4]$. Used in industrial synthesis of linear, high-density polyethylene (HDPE) under low pressure.

4. Composite Materials & FRP

Composites are multiphase materials created by combining two physically distinct materials: Matrix Phase (continuous binder) + Reinforcement Phase (fibers/particles providing high strength).

Fiber-Reinforced Plastics (FRP):

- Glass Fiber Reinforced Plastic (GFRP): High tensile strength, lightweight, corrosion resistant. Used in boat hulls, water tanks, automotive body panels.
- Carbon Fiber Reinforced Plastic (CFRP): Extreme rigidity and ultralight weight. Used in aerospace (aircraft wings), racing cars, sports equipment.

5. Quick Exam Revision Summary

- Thermoplastics: Soften on heat (PVC) vs Thermosets: Cross-linked, permanent (Bakelite).
- Conducting Polymers: Conjugated π -bonds (Polyaniline).
- Grignard Reagent: $R-Mg-X$ in dry ether.
- Ziegler-Natta Catalyst: $Al(C_2H_5)_3 + TiCl_4$.